Smart bus tracking and monitoring system using camera

Zeroth Review Presentation

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Introduction

- This project presents a mobile-based bus attendance system that uses face recognition and GPS to track students boarding and exiting the college bus.
- The solution is cost-effective, easy to deploy, and requires only a single smartphone without any additional hardware.

Objective

- Develop an intelligent college bus surveillance and attendance system.
- Ensure accurate real-time student identification using face recognition.
- Track live bus location using GPS.
- Centralized dashboard for unified data storage and monitoring.
- Minimize hardware cost using smartphones instead of special devices.

Existing System

- Manual attendance marking in buses leads to delays and errors.
- RFID or card-tap systems depend on students carrying cards.
- Additional hardware deployment increases cost and maintenance.
- No integrated real-time tracking and attendance solution available.

Proposed System

- Automatic attendance using facial recognition.
- Live tracking of college buses.
- Centralized storage of attendance and location data.
- Easy monitoring for administrators.
- Reduces manual work and errors.

Base Paper Details

Base Paper 1:
Bus Attendance System Through Facial Recognition by Implementing AI Integration
Authors: Johnson C, Karthick M, Sakthivel P, Sasikala K, Santhi A (2023)

Base Paper 2:
Smart College Bus Tracker: Real-Time Location and ETA Predictor
Authors: M. Sowndharya, Yuvaraj K, Rahul S, Srichandiran S, Sriram S (2025)

Time Plan

Week	Work
1	Requirement analysis, architecture, module identification, UI wireframing
2	DFD/ER, Flask backend setup, authentication, REST API
3	Web dashboard: admin login, student and bus management
4	Flutter app UI, camera preview, ML Kit detection
5	TensorFlow Lite recognition tuning and accuracy checks
6	Attendance workflow: GPS + timestamp, offline sync
7	Real-time GPS tracking, map integration (Google Maps/OSM)
8	Integration, testing, low-light evaluation, optimization
9	UI polishing, security, performance tuning
10	LaTeX report preparation

References

1. IJRPR, Vol. 8, Issue 11, 2023 – Facial Recognition Attendance System.
2. IJSRED, Vol. 8, Issue 3, 2025 – Smart Bus Tracking and ETA Predictor.
3. Google ML Kit Documentation.
4. TensorFlow Lite Developer Guide.
5. Google Maps OpenStreetMap API Documentation.